ShodweStay Hospitality Dashboard

The main objective of this project is to analyze key performance indicators such as revenue, occupancy rate, Average Daily Rate (ADR), Revenue per Available Room (RevPAR), cancellation rate, and booking patterns to generate meaningful insights and support effective hotel management.

Project Overview

Turning hotel booking data into practical business decisions

Business objective

Measure hotel performance and identify opportunities to improve revenue, occupancy and guest experience.

Tools used

Excel for cleaning and exploration; Power BI for interactive reporting; Tableau for visual analysis.

Decision focus

Understand which cities, room classes and booking channels contribute to results.

Core questions

- How much revenue is generated and realized?
- What are the occupancy, ADR, RevPAR and cancellation rates?
- Which properties, room categories and channels drive bookings?
- Where should hotel managers focus improvement efforts?

Dataset and Key Performance Indicators

A consistent KPI definition keeps all three tools aligned

Total Bookings

Count of booking records

Realized Revenue

Revenue retained/realized after applicable adjustments

Occupancy Rate

Occupied room nights ÷ available room nights

ADR

Room revenue ÷ rooms sold

RevPAR

Room revenue ÷ available rooms

Cancellation Rate

Cancelled bookings ÷ total bookings

Recommended data fields

Booking ID • City • Property • Booking date • Check-in/check-out • Room class • Booking platform • Booking status • Revenue • Rating

Excel: Data Preparation and First-Level Analysis

Use Excel to validate the dataset before building dashboards

- Remove duplicate booking IDs and review missing or inconsistent values.
- Standardize city, room-class, booking-platform and booking-status labels.
- Use formulas or Power Query to calculate revenue, occupancy and cancellation metrics.
- Build PivotTables and PivotCharts for quick summaries by city, month and room class.
- Add slicers and timelines so users can filter the report interactively.

Useful Excel features

Power Query • PivotTables • PivotCharts • SUMIFS/COUNTIFS • XLOOKUP • Slicers

Quality checks

Check duplicates, blank values, date ranges, negative revenue and inconsistent status labels.

Output

A clean, documented dataset and a refreshable summary report ready for BI tools.

Hospitality Analytics BI Suite

INTERACTIVE SLICERS — LIVE 3-TIER

SELECT CITY — 4 CITIES

- Bangalore
- Delhi
- Hyderabad
- Mumbai

ROOM CATEGORY — 4 CLASSES

- RT1
- RT2
- RT3
- RT4

OPERATING MONTH — MAY - JUL

- Jul-22
- Jun-22
- May-22

LIVE DATA MODEL LINKED

Cross-filtering active across all 4 charts

25 Properties · ₹1.71B Revenue

AdventureWorks 0-Z Slicer Architecture

EXECUTIVE COMMAND CENTER - MASTER OVERVIEW

HOSPITALITY LUXURY BI SUITE · REALIZED CASHFLOW & ASSET UTILIZATION

MAY - JULY 2022 · W19 - W32 | LIVE MODEL LINKED | EXECUTIVE BI

REALIZED REVENUE (TARGET MET)	OCCUPANCY % (+2.8% WoW)	AVERAGE DAILY RATE (HIGH YIELD)	REVPAR (YIELD) (INDEX 1.14)	REALISATION RATE (OPTIMAL 70%+)
₹1,708.8M	57.87%	₹12,696	₹7,347	70.15%
Net Realized Revenue	Capacity Utilisation	ADR per Sold Room	RevPAR per Available Room	Checked Out Stays %

WEEKLY REVENUE & REALISATION TREND

W19 - W32 Line Chart

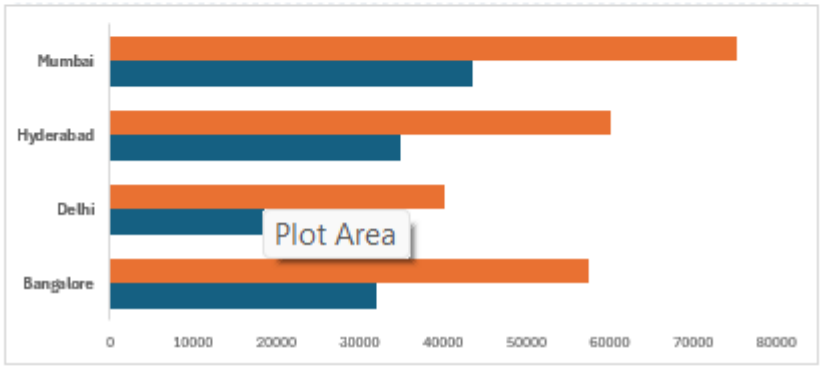
14-week revenue generation and cash realisation stability



CITY PERFORMANCE MATRIX & SHARE

Ranked Bar Chart

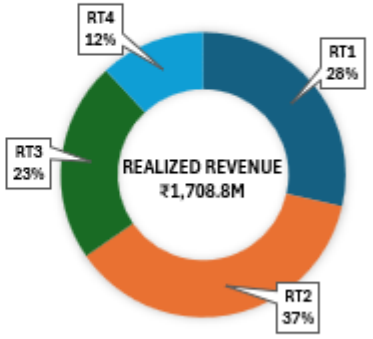
Mumbai (₹668.6M) · Bangalore (₹420.4M) · Hyd (₹325.2M) · Delhi (₹294.5M) — Revenue — Bookings



ROOM CLASS PERFORMANCE & YIELD MIX

Donut Chart

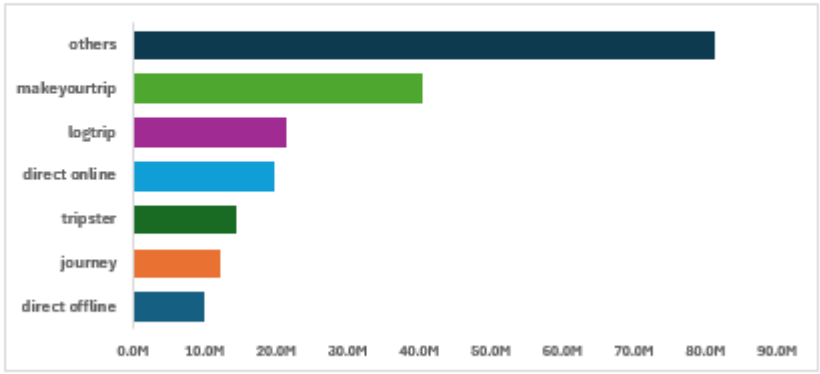
Elite (₹560M) · Premium (₹462M) · Presidential (₹377M) · Standard (₹310M)



BOOKING PLATFORM CONTRIBUTION

Horizontal Bar Chart

Others (40.9%) · MakeMyTrip (19.8%) · LogTrip (10.9%) · Direct Online (9.0%)

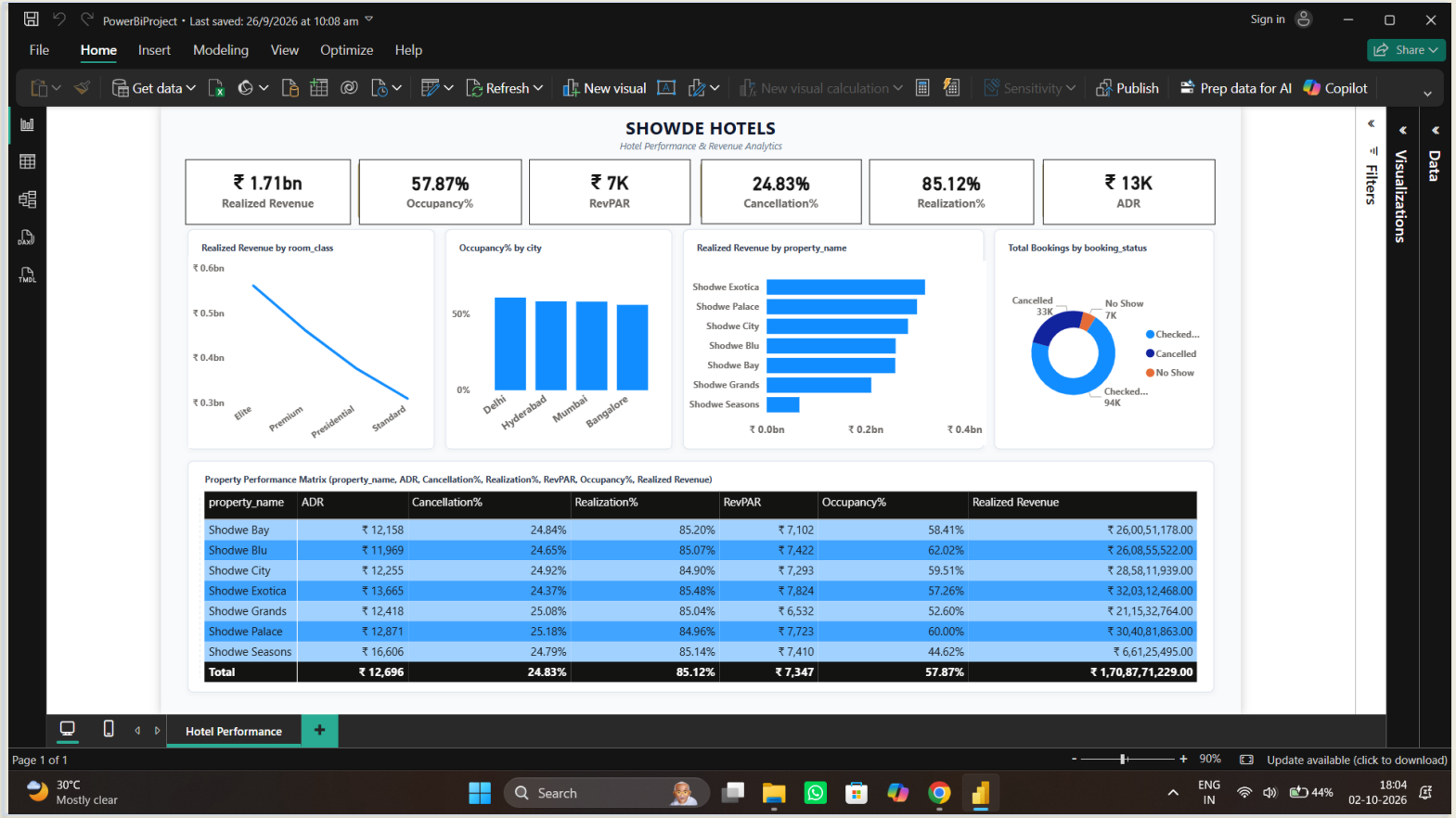


Shodwe Hotel Luxury BI Suite · Active Enterprise Architecture · Powered by Power Query & Excel Data Model

Luxury Light Edition · MASTER OVERVIEW

Power BI: Interactive Hotel Performance Dashboard

Executive view of revenue, occupancy, cancellation and property performance



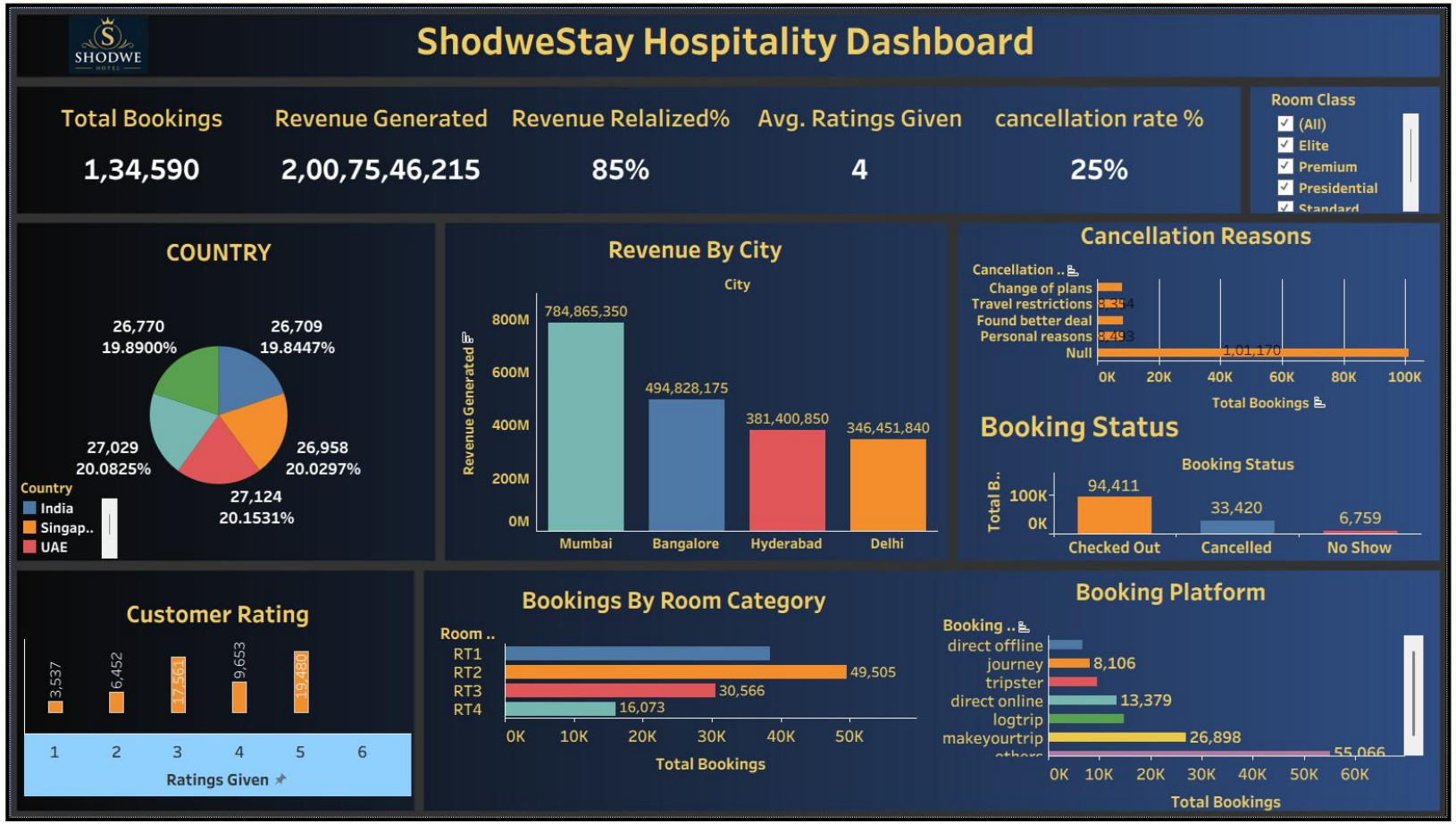
What the dashboard communicates

- KPI cards summarize headline performance.
- Charts compare room classes, cities and properties.
- Booking-status visuals show checked-out, cancelled and no-show patterns.
- The property matrix supports side-by-side operational comparison.

Example screenshot supplied for this project

Tableau: Visual Exploration and Performance Comparison

Explore patterns with filters, comparisons and focused views



Suggested Tableau views

- Weekly revenue and realization trend.
- City-level revenue and booking comparison.
- Room-class performance and revenue mix.
- Booking-platform contribution.
- Interactive city, room category and month filters.

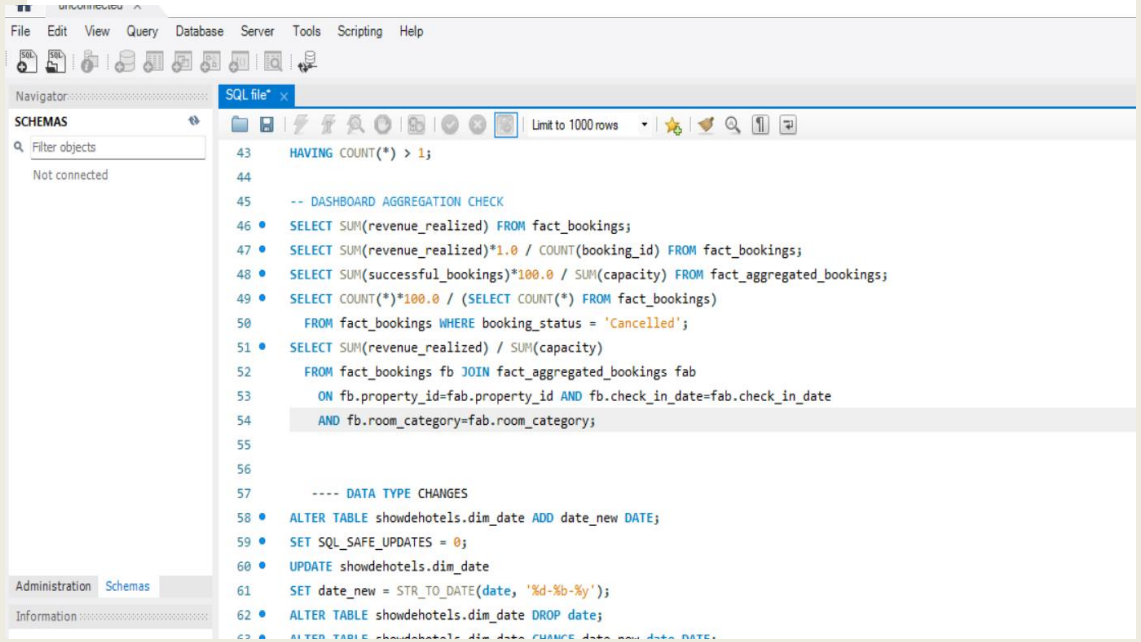
Reference dashboard mock-up

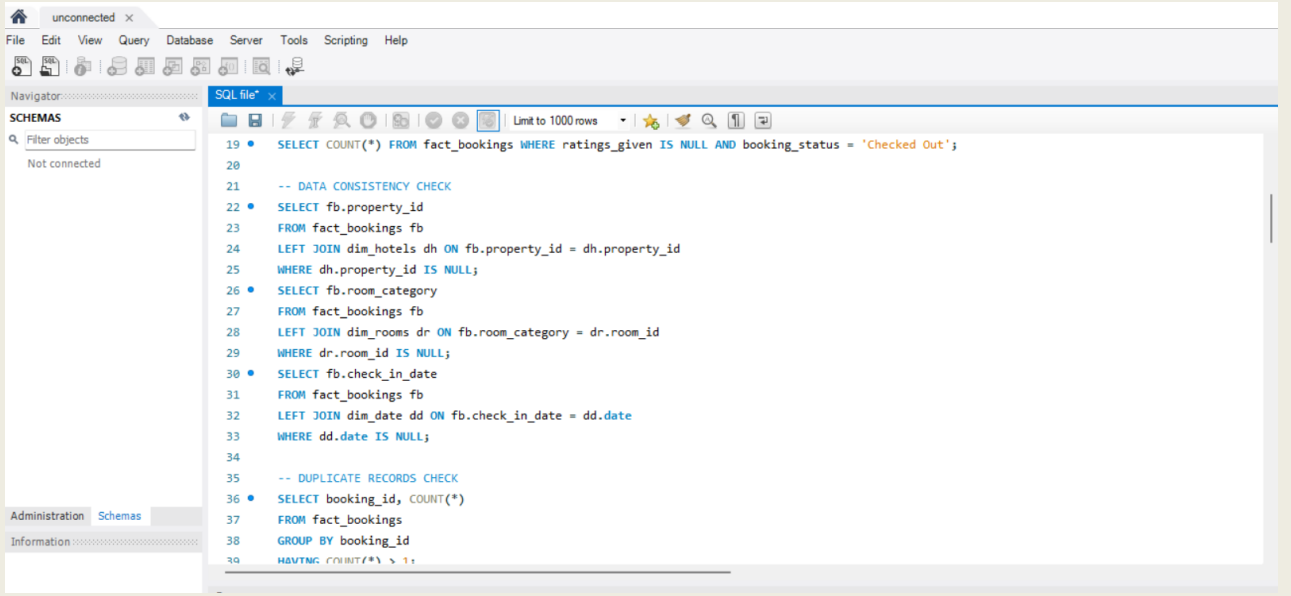
SQL Query

```sql
Create database showdehotels;
use showdehotels;
show tables;

-- DATA COUNT VALIDATION
SELECT COUNT(*) FROM dim_hotels;
SELECT COUNT(*) FROM dim_rooms;
SELECT COUNT(*) FROM dim_date;
SELECT COUNT(*) FROM fact_bookings;
SELECT COUNT(*) FROM fact_aggregated_bookings;

-- DATA COMPLETENESS CHECK
SELECT * FROM fact_bookings WHERE property_id IS NULL OR check_in_date IS NULL OR room_category IS NULL;
SELECT * FROM fact_bookings WHERE booking_status NOT IN ('Checked Out','Cancelled','No Show');
SELECT COUNT(*) FROM fact_bookings WHERE ratings_given IS NULL AND booking_status = 'Checked Out';
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Insights Gained from the Project

After analyzing the hospitality dataset using Excel, Power BI, and Tableau, we gained valuable insights into hotel revenue, occupancy, booking patterns, cancellation rates, and property performance. The analysis showed an overall occupancy rate of 57.87%, a realized revenue of approximately ₹1.71 billion, and a cancellation rate of 24.83%. The city-wise analysis indicated that Mumbai generated the highest revenue among the four cities displayed, followed by Bangalore, Hyderabad, and Delhi. We also identified differences in occupancy and revenue across hotel properties, variations in room-category bookings, and differences in booking platform contributions. These insights help hotel management identify performance gaps, improve room utilization, manage cancellations, and make data-driven decisions to increase operational efficiency and revenue.

How to Interpret the Results

Use the visuals to investigate drivers, not just report totals

Revenue by city

Compare both total revenue and booking volume. High revenue may reflect more bookings, higher room rates or both.

Room-class mix

Review booking share, average rate and occupancy together to understand which room categories create value.

Cancellation patterns

Break down cancellations by platform, lead time, property and month to locate potential friction points.

Guest ratings

Compare ratings with property, room class and booking channel. Check sample sizes before drawing conclusions.

Booking platforms

Measure booking contribution and cancellation rate by channel, not only total booking count.

Time trends

Look for seasonality and sudden changes; compare equivalent periods where possible.

Important: the screenshots are visual references. Confirm all figures against the underlying dataset before presenting findings as final results.

Business Recommendations

Translate dashboard findings into actions that can be measured

Revenue management

Review ADR and RevPAR by property and room class. Test rate adjustments against occupancy and demand.

Cancellation control

Identify high-cancellation channels and booking windows. Test clearer policies and reminder communications.

Channel strategy

Compare net revenue and cancellation rates across platforms before changing channel allocation.

Property operations

Use occupancy forecasts to align staffing, housekeeping and room availability.

Guest experience

Track ratings by property and issue type, then monitor whether service changes improve scores.

Conclusion

From raw booking records to actionable hospitality insights

The project demonstrates how Excel, Power BI and Tableau can be used together to analyze hospitality data. Excel supports data preparation and initial analysis, Power BI provides an interactive management dashboard, and Tableau enables visual exploration of hotel performance.
By combining accurate data, consistent KPIs and meaningful visualizations, hotel managers can better understand revenue, occupancy, cancellations and booking patterns to support informed operational and commercial decisions.

Thank you | Questions?